

Lecture 18 : MLE - Asymptotic Properties. (Large sample properties)

Defn. (Recall) An estimator $\delta_n(\underline{x})$ of $g(\theta)$ is said to be consistent (for $g(\theta)$) if $\delta_n(\underline{x}) \xrightarrow{P} g(\theta)$ as $n \rightarrow \infty$.

(Use Defn of convergence in probability, the Chebyshev's inequality or the WLLN to prove consistency).

Sufficient ('if') conditions of consistency.

$\delta_n(\underline{x})$ is consistent for $g(\theta)$ if any one of the following equivalent conditions holds:

- i) $MSE_{\theta}(\delta_n(\underline{x})) = E_{\theta}(\delta_n(\underline{x}) - g(\theta))^2 \rightarrow 0$ as $n \rightarrow \infty$, $\forall \theta \in \Theta$.
- ii) $Bias_{\theta}(\delta_n(\underline{x})) \rightarrow 0$ and $Var_{\theta}(\delta_n(\underline{x})) \rightarrow 0$ as $n \rightarrow \infty$, $\forall \theta \in \Theta$.
- iii) $E_{\theta}(\delta_n(\underline{x})) = g(\theta) \forall \theta \in \Theta$ and $Var_{\theta}(\delta_n(\underline{x})) \rightarrow 0$ as $n \rightarrow \infty$.

Property 1 If $\hat{\theta}_n$ is MLE of the true parameter θ_0 (based on n observations), then Consistency of $\hat{\theta}_n \xrightarrow{P} \theta_0$ as $n \rightarrow \infty$.

MLE By invariance property (holds for small-sample also) of MLE, $g(\hat{\theta}_n) \xrightarrow{P} g(\theta_0)$ as $n \rightarrow \infty$, for any real fn $g(\cdot)$.

i.e., MLE is a consistent estimator of the corresponding parameter

Example $X_1, \dots, X_n$ iid $N(\theta, \sigma^2)$. Then,

$\delta_{1n}(\underline{x}) = \bar{X}_n^2 - \frac{\sigma^2}{n}$ is the UMVUE of $g(\theta) = \theta^2$ if σ^2 is known.

$$\delta_{2n}(\underline{x}) = \bar{X}_n^2 - \frac{s^2}{n}$$

unknown

$$\delta_{3n}(\underline{x}) = \bar{X}_n^2$$

in the MLE of θ^2 , " " irrespective of whether σ^2 is known or unknown.

i) All are consistent estimators of θ^2

ii) $\delta_{1n}(\underline{x})$ & $\delta_{2n}(\underline{x})$ are unbiased estimators of θ^2

iii) $\delta_{3n}(\underline{x})$ is asymptotically unbiased (as $n \rightarrow \infty$)

Check

2. Asymptotic Normality of MLE under the following regularity conditions:

2/18

(i) $X_1, \dots, X_n, \dots$ are random observations from $\theta \in \mathcal{P}_\theta, \theta \in \mathcal{P}$. The family of distns $\mathcal{P}_\theta, \theta \in \mathcal{P}$, have a common support.

(ii) The parameter space $\mathcal{P}$ contains an open interval, and θ_0 , the true parameter is an interior point of this interval.

Under regularity conditions, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{L} N(0, \frac{1}{I(\theta_0)})$ as $n \rightarrow \infty$

where $I(\theta_0)$ is the Fisher info' an unbiased estimator of $g(\theta)$ at θ_0 .

For the multiparameter case, $\theta_{p \times 1} \in \mathcal{P} \subseteq \mathbb{R}^p$.

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{L} N_p(0, (I(\theta_0))^{-1})$, $I(\theta_0)$ is the Fisher information matrix at θ_0 loc.

3. Asymptotic Efficiency of MLE

An efficient estimator (in the finite sample sense) is an unbiased estimator whose variance attains the CRLB.

(All efficient estimators are UMVUEs of the corresponding parameters, but all UMVUEs are not efficient estimators.)

$\hat{\theta}_n$ is an asymptotically efficient estimator of $g(\theta) = \theta$, $\therefore$ ~~since~~ the asymptotic var. $\frac{1}{I(\theta)}$ is equal to the CRLB.

Example (includes Property 2 & 3)

$X_1, \dots, X_n, \dots$ (equivalently, $\{X_n\}_{n=1}^\infty$) are iid $P(\theta)$, $\hat{\theta}_n = \bar{X}_n$, ^{MLE} θ based on n obsns.

Then, $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \frac{1}{I(\theta)})$

Note, $I(\theta) = E\left[\frac{\partial}{\partial \theta} \ln f_\theta(x)\right]^2 = -E\left[\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x)\right] = \frac{1}{\theta}$.

Hence, $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \theta)$

(Recall, $V(\bar{X}_n) \geq \frac{(g'(\theta))^2}{nI(\theta)}$ (with $g(\theta) = \theta$)

$= \frac{\theta}{n}$, the CRLB of $\bar{X}_n$, and $\bar{X}_n$ is asymptotically efficient.

By invariance property of MLE & Δ -method, we also have

$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{L} N(0, \frac{(g'(\theta))^2}{I(\theta)})$

i.e., var of $g(\hat{\theta}_n) = \frac{(g'(\theta))^2}{nI(\theta)}$, giving asymptotic efficiency of $g(\hat{\theta}_n)$